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To the Editor-in-Chief
IEEE Transactions on Information Theory

Dear Editor-in-Chief,

I am pleased to submit the enclosed manuscript, “Observation Theory: Geometry, Distortion, and Reliability as Properties of Observation,” for consideration as a regular paper in the *IEEE Transactions on Information Theory*.

The paper develops a consumer-relative theory of lossy representation. From a second-order expansion of a downstream task’s distortion it isolates a single controlling quantity—the reconstruction error read through the consumer’s read operator $P_{\mathcal{C}} = J^{\top} G J$ —and shows that ordinary mean-squared-error reconstruction is its isotropic corner $P_{\mathcal{C}} = I$, the corner in which Shannon’s Gaussian rate–distortion function and Gauss least squares reside. The principal contributions are: (i) an identification of the read operator with the pullback geometry of *distinguishability* and the quotient of the representation it induces; (ii) the *identifiability* of the read operator from black-box consumer queries, validated blind against a planted ground truth; (iii) a consumer-relative rate–distortion function with a proof of achievability and converse for a general source; and (iv) a coding theorem for the *true* nonlinear consumer divergence that reduces exactly to output-side coding, together with its high-resolution form. The framework is positioned as the identifiable geometric middle term between Shannon rate–distortion and the information bottleneck.

The work lies squarely within the Transactions’ scope—Shannon theory, rate–distortion and source coding, quantization, and information geometry, with connections to learning and inference. It draws on and extends the indirect/remote source-coding tradition (Dobrushin–Tsybakov, Wolf–Ziv, Witsenhausen), coding for computing (Körner, Orlitsky–Roche), and the rate–distortion–perception and information-bottleneck literatures, all of which are cited and discussed.

The material is original and previously unpublished; it has not been presented, in whole or in part, at any conference, and it is not under review at any other journal. It relates to, but is distinct from, two companion works by the author (on the spectral/angular observer and on geometric key quantization) that are cited in the manuscript; the present paper is a self-contained information-theoretic synthesis and is not a verbatim extension of either.

In the interest of reproducible research, every empirical claim in the paper is governed by a sealed, before-the-fact registration with a pre-declared acceptance bar. The pre-registrations, experiment code, committed result artifacts, and a continuous-integration workflow that re-runs the reproducible experiments and verifies the paper’s theoretical constructions are publicly available at the repository named in the manuscript. Consistent with the Society’s encouragement, I intend to post a preprint to arXiv at the time of submission.

I confirm that I am the sole author; that the submission is prepared in the required single-column format and is well within the 50-page review limit; and that I hold a registered ORCID. I would

be glad to suggest qualified reviewers or to provide any additional material the editors may find helpful.

Thank you for your time and for considering this work.

Sincerely,

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